ANNEXURE -7.4

STANDARD OPERATING PROCEDURE (SOP)

Sr.No.	Focus Area	SOP Description
1.	**Loading and Unloading**	**1. Prior to loading/unloading** ▪ Visually check all hoses for leaks and wet spots. ▪ Verify that sufficient volume is available in the storage tank or truck. ▪ Secure the tank vehicle with wheel chocks and interlocks. ▪ Verify that the vehicle's parking brakes are set. ▪ Verify proper alignment of valves and proper functioning of the pumping system. ▪ Establish adequate bonding/grounding prior to connecting to the fuel transfer point. **2. During Loading/Unloading** ▪ Driver must stay with the vehicle at all times during loading/unloading activities. ▪ Facility manager or designee should observe the delivery driver during loading/unloading. ▪ Periodically inspect all systems, hoses and connections. ▪ When loading, keep internal and external valves on the receiving tank open along with the pressure relief valves. ▪ When making a connection, shut off the vehicle engine. When transferring class 3 materials, shut off the vehicle engine unless it is used to operate a pump. ▪ Maintain communication with the pumping and receiving stations. ▪ Monitor the liquid level in the receiving tank to prevent overflow. ▪ Monitor flow meters to determine rate of flow. ▪ When topping off the tank, reduce flow rate to prevent overflow. **3. Loading and Unloading** ▪ Driver must stay with the vehicle at all times during loading/unloading activities. ▪ Facility manager or designee should observe the delivery driver during loading/unloading. ▪ Periodically inspect all systems, hoses and connections. ▪ When loading, keep internal and external valves on the receiving tank open along with the pressure relief valves.

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		▪ When making a connection, shut off the vehicle engine. ▪ Maintain communication with the pumping and receiving stations. ▪ Monitor the liquid level in the receiving tank to prevent overflow. ▪ Monitor flow meters to determine rate of flow. ▪ When topping off the tank, reduce flow rate to prevent overflow. **4. After Loading/Unloading** ▪ Make sure the transfer operation is completed. ▪ Close all tank and loading valves before disconnecting. ▪ Securely close all vehicle internal, external, and dome cover valves before disconnecting. ▪ Secure all hatches. ▪ Disconnect grounding/bonding wires. ▪ Make sure the hoses are drained to remove the remaining oil before moving them away from the connection. Use a drip pan. ▪ Cap the end of the hose and other connecting devices before moving them to prevent uncontrolled leakage. ▪ Remove wheel chocks and interlocks. ▪ Inspect the lowermost drain and all outlets on tank truck prior to departure. If necessary, tighten, adjust, or replace caps, valves, or other equipment to prevent leaking while in transit.
2.	**SOP for Reactor:**	▪ Select a vessel which has the capacity, pressure rating, corrosion resistance and design features that are suitable for its intended use. ▪ Operate the vessel within a suitable safety devices and controllers ▪ Establish training procedures to ensure that any person handling the equipment knows how to use it properly. ▪ Maintain the equipment in good condition, and test periodically per the vendor's instructions to ensure that the vessel is remains structurally sound. ▪ Complete a hazard assessment before initiating the experiment, including**:** o Assessment of any intermediates, side-products and products that may form and their behavior within the vessel, including their corrosive nature and their tendency to violently decompose at elevated temperature and pressure. o Determination of maximum temperature and pressure limits expected, taking into

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		account the energetics of the reaction being conducted and any pathways that might cause the reaction to run out of control. ▪ Maintain adequate ventilation. This can be achieved by installing the reactor within a fume hood, attaching tubing to the rupture disk that extends to an appropriate exhaust such as the interior of a fume hood, or by ensuring that the lab area as a whole has adequate ventilation and that the reactor is installed near an exhaust fan (in the case of larger reactors). ▪ Run preliminary experiments using small quantities of reactants when starting work with new or unfamiliar materials. ▪ Use appropriate PPE, including safety glasses, chemical resistant gloves, a lab coat, and also a face shield for operations that present particular hazards. ▪ Keep a log of usage for each vessel. Information on the log should include temperature, pressure, reagents/solvents used, and any inspections and tests it has undergone
3.	**Handling of Chemicals**	▪ Wash thoroughly after handling. ▪ Remove contaminated clothing and wash before reuse. ▪ Ground and bond containers when transferring material (i.e., if in a metal container). For safety reasons, transfer from glass to glass container using proper techniques (often purged with nitrogen). ▪ Never add water to this product. ▪ Avoid shock and friction. ▪ Use spark-proof tools and explosion proof equipment. ▪ Do not get in eyes, on skin, or on clothing. ▪ Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. ▪ Keep container tightly closed. ▪ Do not ingest or inhale. ▪ Do not pressurize, cut, weld, braze, solder, drill, grind, or expose empty containers to heat, sparks or open flames. ▪ Keep away from heat, sparks and flame. ▪ Use only with adequate ventilation or respiratory protection.
4.	**Storage of Chemicals/Raw Materials/Products**	▪ Keep storage area clean and free from leakages and spillages. ▪ Keep storage area dry condition all time. ▪ Keep away from heat, sparks & flame.

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		▪ Keep away from sources of ignition. ▪ Keep away from acids and moisture. ▪ Do not store in direct sunlight or strong incandescent light. ▪ Storage under a nitrogen blanket has been recommended OR inside a glove box (under inert atmosphere). ▪ Isolate from oxidizing materials and acids.
5.	**Spills/Leaks**	▪ Remove all sources of ignition. ▪ Absorb spill using an absorbent, non-combustible material such as earth, sand, or vermiculite. ▪ Do not use combustible materials such as sawdust. ▪ Do not get water inside containers. ▪ Approach spill from upwind (if outdoors). ▪ Use water spray to cool and disperse vapors, protect personnel, and dilute spills to form nonflammable mixtures. ▪ Control runoff and isolate discharged material for proper disposal (if outdoors).

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